

LABORATORY RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 1

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1. TITLE

Design Static Webpage using HTML Components

2. AIM

To design and develop a structured static webpage using HTML5 semantic elements and HTML Components.

3. OBJECTIVE

The objectives of this experiment are:

- To understand the basic structure of an HTML5 document.
 - To create a static webpage using HTML components.
 - To use semantic HTML elements such as <header>, <nav>, <main>, <section>, <article>, <aside> and <footer>.
 - To create navigation links within a webpage.
 - To create a user inquiry form using HTML form controls.
 - To organize webpage information using ordered and unordered lists.
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4. THEORY

HTML (HyperText Markup Language) is the standard markup language used to create and structure web pages. A static webpage contains fixed content that is displayed directly by a web browser.

HTML5 provides semantic elements that clearly describe the purpose of different parts of a webpage. The <header> element is used for the introductory section, <nav> contains navigation links, <main> contains the primary content, <section> groups related content, <article> represents independent content, <aside> contains supplementary information, and <footer> represents the closing section of the webpage.

HTML also provides various components for collecting user input. Form controls such as text boxes, email fields, drop-down lists, text areas, submit buttons and reset buttons can be created using HTML. In this experiment, these components are used to design a complete static webpage using pure HTML elements.

5. ALGORITHM / PROCEDURE

1. Create a new HTML file and declare the HTML5 document type using <!DOCTYPE html>.
 2. Create the <html>, <head> and <body> sections.
 3. Add the webpage title inside the <title> element.
 4. Create a header containing the webpage name and description.
 5. Create a navigation bar containing links to each section of the page.
 6. Add a section demonstrating text formatting elements such as bold, italic, underline, highlight, subscript and superscript.
 7. Add a section demonstrating hyperlinks.
 8. Add a section containing ordered and unordered lists.
 9. Add a section containing a table to display structured data.
 10. Add a section containing an image element.
 11. Add a section containing an iframe to embed another webpage.
 12. Create a user inquiry form using HTML form controls such as text input, email input, select, textarea and submit button.
 13. Create a footer containing copyright information.
 14. Save the HTML file and open it in a web browser.
 15. Verify the webpage structure, navigation links and form controls.
 16. Compare the actual output with the expected output.
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6. PROGRAM

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>HTML Components</title>
</head>
<body>

<header>
  <h1>HTML Components</h1>
  <p>Fundamental Building Blocks of HTML5</p>
</header>

<hr>

<nav>
  <a href="#text">Text Formatting</a> |
  <a href="#links">Hyperlinks</a> |
  <a href="#lists">Lists</a> |
  <a href="#table">Table</a> |
  <a href="#image">Image</a> |
  <a href="#form">Form</a>
</nav>

<hr>

<main>

  <section id="text">
    <h2>1. Text Formatting</h2>

    <p>
      HTML is used to create <strong>structured web pages</strong>.
    </p>

    <p>
      This text shows <b>Bold</b>,
      <i>Italic</i>,
      <u>Underlined</u>,
      <mark>Highlighted</mark>,
      <small>Small Text</small>,
      H<sub>2</sub>O and
      X<sup>2</sup>.
    </p>
  </section>

  <hr>

  <section id="links">
    <h2>2. Hyperlinks</h2>

    <p>
      Visit
      <a href="https://www.w3schools.com/" target="_blank">
        W3Schools
      </a>
      to learn more about HTML.
    </p>
  </section>

  <hr>

  <section id="lists">
    <h2>3. Lists</h2>

    <h3>Unordered List</h3>
    <ul>
      <li>HTML</li>
      <li>CSS</li>
      <li>JavaScript</li>
    </ul>

    <h3>Ordered List</h3>
    <ol>
      <li>Learn HTML</li>
      <li>Practice Programs</li>
      <li>Build Projects</li>
    </ol>
  </section>

  <hr>

  <section id="table">
    <h2>4. Table</h2>

    <table border="1" cellpadding="10">

```

```

        <tr>
            <th>HTML Component</th>
            <th>Purpose</th>
        </tr>

        <tr>
            <td>Heading</td>
            <td>Displays titles and headings</td>
        </tr>

        <tr>
            <td>Paragraph</td>
            <td>Displays text content</td>
        </tr>

        <tr>
            <td>Image</td>
            <td>Displays an image</td>
        </tr>

        <tr>
            <td>Form</td>
            <td>Collects user input</td>
        </tr>
    </table>
</section>

<hr>

<section id="image">
    <h2>5. Image</h2>

    

    <p>This is an example of an HTML image.</p>
</section>

<hr>

<section>
    <h2>6. Iframe</h2>

    <iframe
        src="https://www.example.com"
        width="500"
        height="250"
        title="Example Website">
    </iframe>
</section>

<hr>

<section id="form">
    <h2>7. HTML Form</h2>

    <form>

        <label>Name:</label><br>
        <input type="text"
            placeholder="Enter your name">
        <br><br>

        <label>Email:</label><br>
        <input type="email"
            placeholder="Enter your email">
        <br><br>

        <label>Course:</label><br>
        <select>
            <option>HTML</option>
            <option>CSS</option>
            <option>JavaScript</option>
        </select>
        <br><br>

        <label>Message:</label><br>
        <textarea rows="4"
            cols="40"
            placeholder="Enter your message"></textarea>

        <br><br>

        <input type="submit" value="Submit">

```

```

        </form>
    </section>

</main>

<hr>

<footer>
    <p>&copy; 2026 HTML Components Static Web Page</p>
</footer>

</body>
</html>

```

7. CODE EXPLANATION

`<!DOCTYPE html>` - Declares that the document uses HTML5.

`<html>` - Defines the root element of the HTML document.

`<head>` - Contains metadata and the title of the webpage.

`<body>` - Contains the visible content of the webpage.

`<header>` - Defines the header section of the webpage.

`<nav>` - Contains navigation links such as Text Formatting, Hyperlinks, Lists, Table, Image and Form.

`<section>` - Groups related content into a logical section.

`<main>` - Defines the main content of the webpage.

`<mark>`, `<sub>`, `<sup>` - Used to highlight text and display subscript/superscript characters.

`<table>` - Displays data in a structured tabular format.

`<img>` - Embeds an image into the webpage.

`<iframe>` - Embeds another webpage within the current page.

`<form>` - Creates a form for collecting user information.

`<input>` - Creates different input controls such as text, email and submit fields.

`<select>` - Creates a drop-down list containing different options.

`<textarea>` - Creates a multi-line text input area.

`<ul>` - Creates an unordered list.

`<ol>` - Creates an ordered list.

`<footer>` - Defines the footer section of the webpage.

8. TEST CASES AND EXECUTION DATA

I would not just write three random test cases. I would actually execute them and record the results.

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Open the HTML file in a web browser	Static webpage should be displayed correctly	Static webpage displayed correctly	Pass
TC-02	Click each navigation link (Text Formatting, Hyperlinks, Lists, Table, Image, Form)	Corresponding section should be displayed	Corresponding section displayed	Pass
TC-03	Check text formatting elements (bold, italic, underline, highlight, subscript, superscript)	All formatting styles should render correctly	All formatting styles rendered correctly	Pass
TC-04	Click the W3Schools hyperlink	Link should open in a new tab	Link opened in a new tab	Pass
TC-05	Enter name, email, course and message in the form and submit	Form should accept the entered information	Information accepted	Pass
TC-06	Check the iframe and image elements	Embedded webpage and image should load correctly	Embedded webpage and image loaded correctly	Pass

9. OUTPUT

HTML Components

Fundamental Building Blocks of HTML5

[Text Formatting](#) | [Hyperlinks](#) | [Lists](#) | [Table](#) | [Image](#) | [Form](#)

1. Text Formatting

HTML is used to create **structured web pages**.

This text shows **Bold**, *Italic*, Underlined, Highlighted, Small Text, H₂O and X².

2. Hyperlinks

Visit [W3Schools](#) to learn more about HTML.

3. Lists

Unordered List

- HTML
- CSS
- JavaScript

Ordered List

1. Learn HTML
2. Practice Programs
3. Build Projects

4. Table

HTML Component	Purpose
Heading	Displays titles and headings
Paragraph	Displays text content
Image	Displays an image
Form	Collects user input

5. Image

Sample Image

This is an example of an HTML image.

6. Iframe

Example Domain

This domain is for use in documentation examples without needing permission. Avoid use in operations.

[Learn more](#)

7. HTML Form

Name:

Email:

Course:

HTML

Message:

10. RESULT

Thus, a structured static webpage was successfully designed and developed using HTML5 semantic elements and HTML components. The webpage was verified for its structure, navigation links, form controls, lists, table, image, iframe and footer.